

How Abstraction Causes Information to Lose Meaning

Selection, Compression, Reconstruction, and Recursive Compression

A. Jacobs | Semantic Fidelity Project

Series Context

This paper applies the definitions in *What Is Semantic Fidelity*, *When Accuracy Is Not Enough*, and *The Compression Paradox in AI*. Semantic fidelity concerns the preservation of meaning across transformation. Semantic drift is gradual change in that meaning. Reality Drift is a broader system condition in which representations continue organizing action while losing corrective contact with what they represent.

Shared Illustrative Case

The following report is fictional and isolates the mechanism. A source report says:

“In this small observational sample, Clinic A had fewer repeat visits after introducing remote follow-up. Patients also received additional staffing support. The report cannot determine which change produced the difference, and the result should not be generalized to other clinics without further study.”

A later summary says:

“Remote follow-up reduced repeat visits at Clinic A.”

This paper examines one point at which the later sentence can be produced, accepted, or reused.

Core Claim

Abstraction becomes a semantic fidelity problem when a representation loses relationships required for its intended use, yet continues to be treated as if those relationships survived.

Abstraction makes thought and coordination possible precisely because it leaves things out. Maps omit most of a landscape. Records reduce encounters to fields, while summaries compress reports into forms that can travel into later decisions. These transformations are useful because they discard information. Their quality depends on whether they preserve what the next use requires.

The source report in the illustrative case contains several controlling relationships. Its observation concerns one clinic and a small sample. Remote follow-up was introduced alongside additional staffing. Fewer repeat visits were observed, but the report does not isolate a cause and explicitly limits generalization. The compressed sentence preserves the clinic, intervention, and outcome while

changing the relationship between them. “*After introducing*” becomes “*reduced*.” An observation becomes a causal conclusion.

Meaning can be lost at several distinct steps. **Selection** retains the decline in repeat visits while dropping the staffing change and study limit. **Compression** shortens what remains and removes uncertainty. Through **normalization**, a limited and conditional observation becomes a familiar intervention-outcome statement. **Recontextualization** then places that statement in a planning memo, where it begins to read as evidence for adoption elsewhere. **Reconstruction** supplies the missing causal explanation with the most available one: remote follow-up worked. None of these transformations needs to produce an obviously false or incoherent sentence. The result may actually sound clearer than the source.

What matters is what the representation will later be used to do. “*Remote follow-up was introduced and repeat visits fell*” may be adequate for an index of reported events. It is inadequate as evidence that remote follow-up caused the decline. Semantic fidelity is therefore assessed against both a referent and a purpose. The relevant question is which relationships must survive for a reader to reach a conclusion the source actually permits.

Recursive compression compounds the problem. A summary enters a planning memo, which is condensed into a presentation bullet and later treated as a reported finding. Each version can be accurately copied from the immediately preceding version while moving farther from the original report. Local accuracy against the last representation cannot establish fidelity to the underlying conditions. Once the source is no longer consulted, the missing staffing detail has no routine path back into the account.

Abstraction remains corrigible when its limits stay visible. A compact summary might instead say, “*Repeat visits fell at one clinic after remote follow-up and added staffing; the report could not establish the cause.*” The sentence is still concise, but it preserves the relationship on which later interpretation depends. A source link and a clear statement of the report’s limits also give later readers a way to recover what was omitted when the task changes.

Diagnostic Question

Which omitted relationship would change a reader’s conclusion or action if it were restored?

Connection to Reality Drift

This single compressed sentence is a local fidelity failure. If organizations repeatedly make decisions from such derivative claims while their reviews cannot return to the underlying conditions, the pattern can contribute to Reality Drift. The point of failure is the weakening connection between the representation and the conditions capable of correcting it.